

Port map: what listens where, and who can reach it

Every TCP port the NEC stack binds, grouped by how far it is exposed. Only three ports are open to the public internet; everything else is reachable from the internal network, from the host loopback, or only from inside the Docker network. Each component system can also be pointed at a partner-hosted instance on another IP — those are outbound connections, listed at the end.

● Public internet

● Internal network / gateway

● Host loopback only (127.0.0.1)

● Container network only

● Partner-hosted, another IP

Public internet

Citizens, partners, VHW handsets

Citizens / partners

Browser traffic to the public web interface.

443 HTTPS 80 → redirect

NEC Android app

VHW handsets: FHIR sync, Keycloak login, dispense upload.

443 HTTPS

Admin / operators

SSH into the NEC host for deploy and maintenance.

22 SSH

public access ↓

Edge — API gateway / firewall

nginx + certbot + ufw on the NEC host

nginx reverse proxy

TLS termination; virtual hosts fan out to the internal service ports below. Nothing else needs to be internet-facing.

80 443

ufw firewall

Hardened profile allows only SSH, HTTP and HTTPS inbound. Outbound 443 for ACME certificate renewal.

22 80 443

proxied to internal services ↓

OpenHIM — interoperability layer

Every app transaction enters here and is routed to the mediator.

5001 client HTTP**5000 client HTTPS****8080 admin API****9285 console****NEC mediator**

Orchestrates data flows — fan-out to OpenSRP, LMIS and DHIS2; webhooks; notifications.

3000 mediator API**HAPI FHIR + fhir-proxy**

Clinical record store the Android app syncs against.

8079 proxy**18079 HAPI direct****Keycloak**

Identity provider for app, web and admin logins.

8083**OpenSRP — community health**

Web console plus the legacy OpenSRP server API.

9901 web**9900 server****LMIS — logistics**

OpenLMIS / eLMIS nginx and Reference UI; stock, orders, requisitions.

8082**DHIS2 — health reporting**

Aggregate reporting target for dispense and stock-on-hand data.

8081**NEC web (bkm-web)**

Public web interface served at the site root.

9902**Monitoring**

Grafana dashboards, Prometheus scraping, Loki log aggregation.

3005 Grafana**9290 Prometheus****3100 Loki****Notifications**

SMS / push / email sink for dispense, receipt and low-stock alerts; MailHog captures mail.

8025 MailHog UI**8086 sink**

not reachable off the host ↓

Data tier & internal-only services

Container network (health-net / lmis-net) — no host binding

PostgreSQL

Beneficiary records, FHIR resources, DHIS2, Keycloak, OpenLMIS schemas.

5432

MongoDB

OpenHIM transaction log and configuration.

27017

Redis

OpenSRP server cache and write queue.

6379

Exporters & SMTP

Scraped by Prometheus; MailHog accepts mail from the notification sink.

8080 cAdvisor

9187 pg-exporter

1025 SMTP

optional – outbound to another IP ↓

Partner-hosted systems — another IP

Alternative upstreams; chosen per system, outbound only

National eLMIS

Partner logistics system in place of the in-stack OpenLMIS.

8082 or 443

National DHIS2

Partner reporting instance in place of the in-stack DHIS2.

443 or 8080

Partner OpenSRP

Partner community-health server in place of the in-stack one.

443

Partner FHIR server

External clinical record store in place of the in-stack HAPI FHIR.

443

FULL PORT REGISTER

Binding is what the host actually publishes. 0.0.0.0 means any interface the firewall permits; 127.0.0.1 means the port answers only on the host itself and must be reached through the gateway or an SSH tunnel.

PORT	SERVICE	BINDING	PURPOSE
PUBLIC INTERNET — THE ONLY PORTS OPEN AT THE FIREWALL			
22	SSH	0.0.0.0	Operator access; fail2ban protected
80	nginx HTTP	0.0.0.0	Redirect to HTTPS plus ACME challenge
443	nginx HTTPS	0.0.0.0	All public traffic; TLS terminated here
INTERNAL NETWORK — PUBLISHED ON THE HOST, PROXIED BY THE GATEWAY			
5001	OpenHIM client	0.0.0.0	Transaction entry point for the Android app
8079	fhir-proxy → HAPI FHIR	0.0.0.0	FHIR R4 sync endpoint for the mobile app
8081	DHIS2	0.0.0.0	Health reporting web app and API
8082	OpenLMIS / eLMIS	0.0.0.0	Logistics UI and API (separate compose stack)
8083	Keycloak	0.0.0.0	OIDC login; served under /auth in production
9901	OpenSRP web	0.0.0.0	Community health console (FHIR-backed)
9902	NEC web	0.0.0.0	Public web interface at the site root
9285	OpenHIM console	0.0.0.0	Transaction monitoring and channel admin
3005	Grafana	0.0.0.0	Dashboards over Prometheus and Loki
9290	Prometheus	0.0.0.0	Metrics scraping and query API
8025	MailHog UI	0.0.0.0	Captured outbound mail (sandbox)
HOST LOOPBACK ONLY — NEVER EXPOSED OFF THE MACHINE			
3000	NEC mediator	127.0.0.1	Orchestration API; called via OpenHIM
5000	OpenHIM client HTTPS	127.0.0.1	TLS variant of 5001
8080	OpenHIM admin API	127.0.0.1	Console backend; proxied by nginx over HTTPS
8086	Notification sink	127.0.0.1	SMS / push / email endpoints for the mediator
9900	OpenSRP server	127.0.0.1	Legacy OpenSRP REST API
18079	HAPI FHIR direct	127.0.0.1	Bypasses fhir-proxy; admin and seeding use
3100	Loki	127.0.0.1	Log ingestion from promtail
CONTAINER NETWORK ONLY — NO HOST BINDING AT ALL			
5432	PostgreSQL	health-net	Primary datastore for every service schema
27017	MongoDB	health-net	OpenHIM transactions and config
6379	Redis	health-net	OpenSRP server cache / write queue
3001	Notification sink	health-net	Container port behind host 8086
1025	MailHog SMTP	health-net	Mail submission from the notification sink
9187	postgres-exporter	health-net	Database metrics for Prometheus

PORT	SERVICE	BINDING	PURPOSE
8080	cAdvisor	health-net	Container metrics for Prometheus
9090	Prometheus	health-net	Container port behind host 9290
3000	Grafana / OpenSRP web	health-net	Container ports behind host 3005 / 9901
8080	Keycloak / DHIS2 / OpenSRP / HAPI	health-net	Container ports behind host 8083 / 8081 / 9900 / 18079
80	bkm-web / fhir-proxy / OpenHIM console	health-net	Container ports behind host 9902 / 8079 / 9285
OUTBOUND — ONLY WHEN A SYSTEM POINTS AT A PARTNER INSTANCE			
443	Partner DHIS2 / OpenSRP / FHIR	egress	HTTPS to the partner host on another IP
8082	National eLMIS	egress	Partner LMIS where it is not TLS-fronted
8080	Partner DHIS2	egress	Where the partner exposes DHIS2 directly
443	Let's Encrypt ACME	egress	Certificate issue and renewal (domain deployments)

GATEWAY ROUTING — PUBLIC NAME TO INTERNAL PORT

With a domain in place, nginx terminates TLS on 443 and forwards to the loopback address of each service, so none of the service ports need to be open at the firewall.

PUBLIC ENTRY	FORWARDS TO	SERVICE
/	127.0.0.1:9902	NEC web
/auth/	127.0.0.1:8083	Keycloak
/openhim-api/	127.0.0.1:8080	OpenHIM admin API (HTTPS upstream)
sync....	127.0.0.1:5001	OpenHIM client — app transactions
fhir....	127.0.0.1:8079	HAPI FHIR via fhir-proxy
opensrp....	127.0.0.1:9901	OpenSRP web
lmis....	127.0.0.1:8082	OpenLMIS / eLMIS
dhis2....	127.0.0.1:8081	DHIS2
openhim....	127.0.0.1:9285	OpenHIM console
grafana....	127.0.0.1:3005	Grafana
prometheus....	127.0.0.1:9290	Prometheus (basic auth)
mailhog....	127.0.0.1:8025	MailHog (basic auth)

UPSTREAM ENDPOINTS — OURS OR THEIRS

The middleware reaches each component system through one configured base URL, so any of them can be swapped for a partner-hosted instance on another IP without touching the rest of the stack. The repo default points all six at the in-stack containers.

COMPONENT	SET BY	NEC-HOSTED (IN STACK)	PARTNER-HOSTED (ANOTHER IP)
Logistics / LMIS	OPENLMIS_AUTH_URL	http://auth:8080	http://<their-ip>:8082
	OPENLMIS_MGMT_URL	http://stockmanagement:8080	or https://<their-host>:443
	OPENLMIS_REF_URL	http://referencedata:8080	
Health reporting	DHIS2_URL	http://dhis2-web:8080	https://<their-host>:443 or http://<their-ip>:8080
Community health	OPENSrp_URL	http://opensrp-server:8080	https://<their-host>:443
Clinical records	HAPI_FHIR_URL	http://hapi-fhir:8080/fhir	https://<their-host>:443/fhir

DATA FLOW

- Citizen or VHW reaches the gateway on 443; nginx forwards to the matching internal port.
- App transactions land on OpenHIM 5001, which routes to the mediator on 3000.
- The mediator fans out to OpenSRP, LMIS and DHIS2 and composes the response.
- Only safe-to-show data returns; the public tier never touches raw records.

DEPLOYMENT VARIANTS

- Domain server: firewall allows 22, 80, 443 only; every service port stays on loopback.
- IP-only bring-up with no domain: the service ports listed as internal are opened directly with `ufw allow`, and traffic is plain HTTP.
- Sandbox / laptop: the same ports bind on localhost with no firewall in front.
- Mixed hosting: point any component system at a partner instance on another IP; the in-stack container can then be dropped, freeing its host port.

SOURCE OF TRUTH

- `docker-compose.yml` — every host binding and container port
- `deploy/nginx/bkm.conf`, `subdomains.conf` — gateway routing
- `deploy/setup-server.sh`, `production-installation.md` — firewall rules